5.3

The Cramér–Rao Lower Bound (CRLB)

CRLB Introduction

The CRLB is the minimum possible variance of an unbiased estimator of a parameter θ (or a function, $\tau(\theta)$).
If $\hat{\theta}$ is an unbiased estimator of θ, then the CRLB tells us how small the variance of $\hat{\theta}$ can be.

- This lower bound may not be attainable.
- If an unbiased estimator does attain the associated CRLB, then it has the uniformly minimum variance amoung all unbiased estimators.

5.3.1

Statement of the CRLB and Derivation

The Cramér–Rao Lower Bound

Let $X_1, X_2, \ldots, X_n$ be random variable with joint pdf $f(\vec{x};\theta)$. Assume that the support of the distribution does not depend on θ and that θ is one-dimensional.

Consider estimating some function $\tau(\theta)$ and let $T = t(\vec{X})$ be any unbiased estimator of $\tau(\theta)$.

Then

$$\mathrm{var}[T] \geq \frac{[\tau^{'}(\theta)^2]}{E\left[\left(\frac{\partial}{\partial\theta}\ln f(\vec{X};\theta)\right)\right]},$$

under certain regularity assumptions.

Regularity assumptions for the CRLB

The regularity assumption for the CRLB are:

- $0 < E\left[\left(\frac{\partial}{\partial\theta}\ln f(\vec{X};\theta)\right)\right] < \infty$,
- $\frac{\partial}{\partial\theta}\int f(\vec{x};\theta)d\,\vec{x} = \int \frac{\partial}{\partial\theta} f(\vec{x};\theta)d\,\vec{x}$
- $\frac{\partial}{\partial\theta}\ln f(\vec{x};\theta)$ exists.

CRLB notation

We denote the CRLB by

$$\mathrm{CRLB}_{\tau(\theta)} := \frac{[\tau^{'}(\theta)^2]}{E\left[\left(\frac{\partial}{\partial\theta}\ln f(\vec{X};\theta)\right)^2\right]}.$$

Computation of the CRLB will simplify when the data are i.i.d., but the CRLB holds for more general joint pdfs.

Definition 5.3.1

The Fisher information for a sample is defined as

$$I_n(\theta) = E\left[\left(\frac{\partial}{\partial\theta}\ln f(\vec{X};\theta)\right)^2\right].$$

The Fisher information quantifies the amount of information the sample contains about θ.

We can rewrite the CRLB of θ as

$$\mathrm{CRLB}_{\tau(\theta)} = \frac{[\tau'(\theta)]^2}{I_n(\theta)}.$$

The Cauchy-Schwarz Inequality

For square-integrable functions f and g,

$$\left(\int f(x)g(x)\,dx\right)^2 \leq \left(\int f(x)^2\,dx\right)\left(\int g(x)^2\,dx\right)$$

There are analagous Cauchy-Schwarz inequalities for sums and inner products.

Proof of C-S inequality

Proof of C-S inequality (cont)

Proof of C-S inequality (cont)

Proof of CRLB

Proof of CRLB (cont)

Proof of Equation 5.3.4

Proof of Equation 5.3.4 (cont)

Proof of Equation 5.3.4 (cont)

Proof of Equation 5.3.4 (cont)

Example 5.3.1

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Bernoulli}(p)$.

a. Find the CRLB for the variance of all unbiased estimators of p.

b. Find the CRLB for the variance of all unbiased estimators of $\tau(p) = p(1-p)$.

Example 5.3.1 (cont)

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